Graphic Era HILL UNIVERSITY
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University under section 2(f) of UGC Act, 1956

End Term (Odd) Semester Examination Nov-Dec 2025

Roll no............................

Name of the Program and semester: M.Tech. (CSE), 1st
Name of the Course: Data Warehousing & Data Mining
Course Code: MCS124
Time: 3 hours

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Explain any five critical challenges in Data Warehousing along with possible solution of each challenge. CO1
b. What is OLAP in data warehouse? Explain typical OLAP operations with diagram. CO2
c. What are the differences between three main types of data warehouse usage: information processing, analytical processing and data mining? Explain in detail with example of each. CO2

Q2. (2X10=20 Marks)

a. What is Multi-Dimensional Data Model? Explain with the help of suitable example. CO1
b. Describe the problem of Data Quality with example. Explain the uses of feature subset selection in data preprocessing. CO3
c. Consider the following dataset and find frequent item sets and generate association rules for them. Assume that minimum support threshold (s = 50%) and minimum confident threshold (c = 80%). CO2

Transaction	List of items
T1	I1, I2, I3
T2	I2, I3, I4
T3	I4, I5
T4	I1, I2, I4
T5	I1, I2, I3, I5
T6	I1, I2, I3, I4

Q3. (2X10=20 Marks)

a. Explain the decision tree classification technique with suitable example. CO5
b. What are different classifier evaluation metrics? Explain each with its significance in classification of objects. CO4
c. What is the method to improve the classification accuracy? Discuss in detail with its advantages & disadvantages. CO5

Q4. (2X10=20 Marks)

a. Distinguish between K-Mean and K-Medoids clustering algorithms. Outline various disadvantages of partitioning based clustering algorithms. CO4
b. Explain between Hierarchical based & Density based clustering with one example. CO4
c. Cluster the following eight points (with (x, y) representing locations) into three clusters:
A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9)
Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2).

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The distance function between two points a = (x1, y1) and b = (x2, y2) is defined as-
$P(a, b) = |x2 - x1| + |y2 - y1|$
Use K-Means Algorithm to find the three cluster centers after the second iteration. CO4

Q5. (2X10=20 Marks)

a. What is spatial Data Mining? Explain with suitable example. CO5
b. How the mining of world wide web is different from text mining? Explain in detail. CO5
c. Explain the procedure for generalizing the object Data by using a suitable example. CO4